Defence & vaccination

Keep pathogens out

1 Skin & nose

Skin: physical barrier; antimicrobial substances and clotting at cuts help protect.

Nose: hairs and mucus trap pathogens and particles.

These barriers are **non-specific:** they act against many pathogens.

2 Airways & stomach

Trachea/bronchi: **mucus traps pathogens; cilia move mucus to the throat** for swallowing.

Stomach **hydrochloric acid** kills many swallowed pathogens.

Respond inside

3 White blood cells

Phagocytosis: engulf and digest pathogens.

Antibodies: bind specific pathogen antigens and help destroy the pathogen.

Antitoxins: neutralise bacterial toxins. Different white cells have different roles.

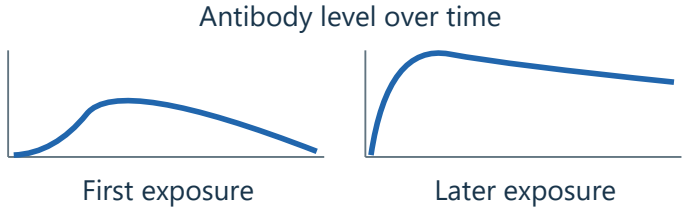
4 Specificity

An antibody's binding site has a shape **complementary to its antigen.**

Antibodies target antigens; antitoxins neutralise toxins. They are not interchangeable terms.

Remember & protect

5 Vaccination sequence



Dead/inactive pathogen or antigens → white cells make specific antibodies → memory cells remain.

Later exposure → **faster, larger antibody response,** destroying pathogens before significant illness.

6 Population protection

High vaccine uptake leaves fewer susceptible hosts, reducing transmission and helping protect those unable to be vaccinated.

Evaluate disease reduction against reliable risk evidence, cost, access and uptake. Protection is not identical for every person.